American College of Rheumatology 2024 Annual Meeting Abstract Summaries

Osteoarthritis

The ACR 2024 Annual Meeting yielded several interesting findings on osteoarthritis (OA). Abstract 0031 demonstrated that intra-articular injection of liraglutide (LIRA), a glucagon-like peptide-1 (GLP-1) analogue, showed significant analgesic, anti-inflammatory, and cartilage-protective effects in a rat model of inflammatory OA, surpassing the effects of dexamethasone. Abstract 0030 highlighted the importance of timing in exercise interventions for OA, finding that exercise was beneficial in early-stage OA but potentially harmful in later stages. This suggests individualized exercise protocols are crucial. Abstract 0023 showcased the effectiveness of an adenosine-guanosine-based oligonucleotide encapsulated in PLGA nanoparticles in reducing cartilage damage and pain in mouse and rat OA models, acting through adenosine receptors A1 and A2A and the AMPK-SIRT1 axis. Abstract 0792 explored the long-term safety and efficacy of lorecivivint, an intra-articular CLK/DYRK inhibitor, showing sustained improvements in pain, function, and medial joint space width, supporting its potential as a disease-modifying OA treatment. Finally, Abstract 0788 reported the long-term maintenance of improvements in pain, stiffness, and function two years after a whole-food plant-based diet, physical activity, and stress management intervention in patients with OA and metabolic syndrome.

Further important findings included research on the role of bone changes in OA. Abstract 0020 investigated the role of CD14 in osteoclast differentiation, finding that CD14-deficient mice showed accelerated osteoclastogenesis and reduced sensitivity to TLR4 inhibition. The identification of novel therapeutic targets for OA was explored in abstract 0021, revealing that CDK1 inhibitors blocked MMP-13 and IL-6 production in both OA chondrocytes and synovial fibroblasts. Research on vagal nerve stimulation (VNS) for OA pain (Abstract 0025) demonstrated its potential to reduce pain and systemic inflammation in two mouse models, suggesting a promising avenue for future clinical trials. Studies on the impact of flexion contracture on OA (Abstract 0022) and the role of menisci in OA development and pain (Abstract 0028) provided further insights into OA pathophysiology. Additionally, several abstracts examined the use of artificial intelligence for OA assessment (L04), novel analgesic agents for OA pain (L08, L15), and the impact of factors like obesity subtypes (0864), sleep quality (0865), and inflammation (1197, 1204) on OA outcomes. Finally, several studies highlighted the increasing global burden of OA and the need for better diagnostic tools and treatment strategies, especially in specific populations and settings (0161, 1880, 2098).

Ankylosing Spondylitis

The ACR 2024 Annual Meeting yielded several interesting abstracts on ankylosing spondylitis (AS). Abstract number 0068 identified GPR65 as a key molecule upregulated in pathogenic Th17 cells and spinal tissue in AS patients, correlating with increased levels of IL-17, IFN-γ, and TNF-α. The finding that GPR65 was highly expressed at sites of ectopic new bone formation in a mouse model further strengthens its potential as a therapeutic target. Abstract 0071 investigated the interleukin (IL)-23/IL-17 pathway, finding that IL-23 may play a more significant role in early axSpA, while the IL-17 pathway appears less dependent on IL-23 in later stages of the disease. This suggests that therapies targeting IL-23 may be more effective in early disease. Abstract 0072 explored the inflammasome's role, suggesting that its activation may be pathogenic in axSpA, with HLA-B27 possibly influencing this mechanism. Abstract 0077 examined late-onset axSpA (>45 years), finding it differed from early-onset forms with shorter disease duration, higher prevalence of psoriasis and fibromyalgia, and greater disease activity. Abstract 0080 investigated the role of aging in AS pathogenesis, identifying NKG2C+CD8+ T cells as cytotoxic effectors potentiated by HLA-B27 stimulation and suggesting NKG2C as a potential therapeutic target. Abstract 0081 examined the impact of the CARD9 S12N mutation on axSpA, revealing a gain-of-function in CARD9 signalling within neutrophils and enhanced neutrophil-mediated Th17 expansion. Abstract 0078 examined the therapeutic potential of TRABID inhibition in AxSpA, with early proof-of-concept for a novel therapeutic target, although further exploration of its cellular and genetic targets is needed. Abstract 0082 compared the IL-23 production in axial versus peripheral entheseal tissue, revealing that axial sites produce significantly more IL-23, suggesting that higher doses of IL-23 blockers might be necessary for effective treatment of AS.

In summary, the ACR 2024 meeting highlighted several promising avenues for AS research and treatment. The identification of GPR65 as a therapeutic target (0068), the potential for stage-specific IL-23/IL-17 pathway modulation (0071), and the role of the inflammasome (0072) offer new directions for drug development. The distinct characteristics of late-onset axSpA (0077), the involvement of aging-associated immune cells (0080), the identification of CARD9 gain of function (0081) and TRABID (0078) as potential therapeutic targets, and the higher IL-23 production in axial sites (0082) provide a more nuanced understanding of the disease. These findings together suggest that personalized therapeutic approaches tailored to disease stage, genetic background, and individual patient characteristics may be crucial for effective axSpA management.

Rheumatoid Arthritis

The American College of Rheumatology 2024 Annual Meeting yielded several interesting findings on rheumatoid arthritis (RA). Research on the mechanisms of myocardial dysfunction in RA highlighted the increased expression and co-localization of malondialdehyde-acetaldehyde (MAA) and citrulline (CIT) in the heart tissues of mice with collagen-induced arthritis (CIA) (Abstract 0032). This suggests a potential role for these post-translational modifications in the pathogenesis of RA-related heart failure, although further investigation is needed to confirm this in humans. Another study used spatial transcriptomics to reveal the heterogeneity of the RA synovium, identifying cellular communities and their interactions, showing that CD34+ fibroblasts co-occurred with macrophages and adipocytes, and that complement played a predicted role in fibroblast activity (Abstract 0036). Further, the study of the role of Dual Specificity Phosphatase 22 (DUSP22) in RA pathogenesis revealed its downregulation in RA patients, potentially influencing UBR2-mediated pathways and contributing to heightened T cell activation (Abstract 0038). Finally, investigations into the fine specificity of anti-citrullinated protein antibodies (ACPA) demonstrated an association between HLA-SE alleles and ACPAs targeting non-glycine cit-motifs (Abstract 0033), providing insights into ACPA epitope spreading and disease pathogenesis.

Importantly, several studies explored novel therapeutic approaches. A bispecific antibody targeting PAD2 and PAD4 enzymes showed potent inhibition of citrullination (Abstract 0039), a key process in RA pathogenesis, representing a potential novel treatment strategy. Additionally, research into the complement pathway revealed the efficacy of suppressing Complement Factor B (CFB) using siRNA to resolve inflammatory arthritis in mice (Abstract 0040). A novel 3D RA synovial tissue model showed promise for testing therapeutic compounds on inflammation and angiogenesis (Abstract 042), while studies on vague nerve stimulation (Abstract L10) and targeting the generation of citrullinated autoantigens (Abstract 0039) indicated potential clinical benefits. The discovery of increased prevalence of clonal haematopoiesis (CHIP) in RA patients (Abstract 0037) may also shed light on disease mechanisms and potential intervention points. Finally, several studies leveraged the utility of various biomarkers to help assess disease activity and to predict outcomes. This included studies on serum zonulin and antibody-secreting cells (Abstract 0012), plasma calprotectin (Abstract 0468), and the development of predictive models using machine learning approaches and data from actigraphy (Abstract 0456) and serum metabolomics (Abstract 0485).

Other

The ACR 2024 Annual Meeting yielded several interesting abstracts on the topic of "Other," encompassing diverse areas of rheumatology research. Significant findings included the discovery of novel therapeutic targets for enhancing bone health (Abstract 0027), demonstrating the potential of TRPC1 inhibition in mesenchymal stem cells to increase osteoblast numbers and bone mass. Additionally, research on allogeneic CAR T cell therapies progressed substantially (Abstract 0010, 0008), highlighting the development of cyclosporine/voclosporin-resistant CAR T cells through CRISPR-Cas9 editing of the cyclophilin A gene (PPIA), paving the way for compatibility with established immunosuppressive therapies and improved allogeneic compatibility. In vivo CAR T cell technologies showed great promise (Abstract 0019), with a novel CD8-targeted lipid nanoparticle (tLNP) system (CPTX2309) mediating efficient anti-CD19 CAR mRNA delivery, achieving rapid and durable B cell depletion. Finally, studies investigating the role of type I interferons in SLE (Abstract 0093) revealed a link between chronic epidermal interferon-kappa overexpression and systemic T cell responses to UV exposure, potentially explaining UV-driven disease flares.

Furthermore, the meeting highlighted important health equity and access issues, with several abstracts describing innovative approaches to address these challenges. The creation of ChronicHue, a global online community for BIPOC individuals living with rheumatic disease and chronic illness (Abstract PP12), provided a supportive environment for addressing healthcare disparities and fostering a sense of belonging. Research on improving access to care focused on implementing diverse approaches including improved referral pathways, electronic consults and telemedicine (Abstracts 0201, 1080, 1928), and the use of electronic health records (EHRs) for better patient phenotyping (Abstract 0137) and enhanced medication adherence interventions (Abstract 1053). The implementation of novel diagnostic tools such as infrared thermography (Abstract 0201) and machine learning algorithms for risk stratification (Abstract 0984) was explored as methods for improving patient triage and resource utilization. Finally, studies explored the use of Artificial Intelligence (Abstract 1076, 1740, 2183, 2587) to help address several rheumatology issues including patient education, diagnosis and clinical management decision support.

Vasculitis

The American College of Rheumatology 2024 Annual Meeting yielded several interesting findings on vasculitis. Research on Behçet's syndrome (BS) highlighted the effectiveness of colchicine in managing oral ulcers and other mucocutaneous manifestations. Abstract L14 detailed a randomized controlled trial (RCT) showing significantly fewer oral ulcers in patients who continued colchicine compared to those who discontinued it, supporting its efficacy in managing mucocutaneous and joint involvement in BS. Further investigation into the role of IgM antiphospholipid antibodies (aPL) in antiphospholipid syndrome (APS) was presented in abstract 0108. This study suggests a pathogenic role for isolated IgM aPL, particularly in relation to venous thrombotic events, obstetric complications, and central nervous system lesions. Additionally, abstract 0717 examined the long-term mortality in ANCA-associated vasculitis (AAV), finding that more recent diagnosis was associated with lower 5-year mortality rates, although this did not vary by age at diagnosis.

Several abstracts explored novel aspects of AAV and its management. Abstract 0262 highlighted the significant overlap between *Bartonella henselae* endocarditis (BE) and AAV, with a high prevalence of ANCA positivity and pauci-immune glomerulonephritis in BE patients. Abstract 0725 investigated serum glial fibrillary acidic protein (GFAP) as a potential biomarker for AAV activity, particularly in relation to renal involvement and progression to end-stage kidney disease (ESKD). Research on treatment strategies focused on the efficacy of various agents in specific vasculitides. Abstract 0765 examined the transcriptomic profile of neutrophils in VEXAS syndrome, revealing increased transcription of genes related to inflammation and immune response. The impact of COVID-19 on AAV patients was explored in abstract 0716, which found similar in-hospital mortality and organ-specific complications to the general COVID-19 population, but a higher likelihood of ICU admission. Finally, abstract 0823 evaluated abatacept as an adjunctive therapy for relapsing GPA, finding no significant benefit over placebo. These studies point to the need for continued research to refine diagnostic tools, identify biomarkers, and optimize treatment strategies for AAV and other forms of vasculitis.

Systemic Lupus Erythematosus

The ACR 2024 Annual Meeting yielded several interesting and important findings related to Systemic Lupus Erythematosus (SLE). Research on novel CAR T-cell therapies showed promise. Abstract #0011 explored the Ingenui-T platform, demonstrating successful CAR T-cell manufacturing using whole blood from SLE patients, maintaining less differentiated T cells and achieving CD19-specific cytolysis comparable to healthy donors, although with reduced IFNγ secretion. Abstract #0018 presented preclinical and clinical data supporting the evaluation of CB-010, an off-the-shelf allogeneic CAR-T cell therapy, in a Phase 1 trial for refractory lupus nephritis and extrarenal lupus. This allogeneic approach offers advantages over autologous therapies by streamlining logistics and manufacturing. Abstract #L17 evaluated allogeneic CD19-targeting CAR-NK cell therapy, showcasing its excellent safety and capacity to achieve rapid and long-lasting remission in relapsed and refractory SLE patients. Longitudinal studies, such as abstract #L13, which compared anifrolumab to standard of care, suggested that long-term anifrolumab may delay the onset and reduce the rate of irreversible organ damage. Finally, several abstracts addressed the use of biomarkers and machine learning for the early diagnosis and prediction of outcomes including flares (0094, 0612, 0613, 0630) and damage (0621).

Several studies investigated the heterogeneity of SLE. Abstract #L03 identified a novel CD9-expressing Tfh cell population as the primary producer of IL-21 and IFN-γ, crucial for driving B-cell differentiation into autoantibody-secreting cells. Abstract #0095 highlighted expanded exhausted CD8+ T cells and IL-17 producing gamma-delta T cells in the brains of NPSLE-prone mice, potentially contributing to neuropsychiatric symptoms. Furthermore, research using proteomic and metabolomic analyses (0099, 0100, 0101, 0104, 0109, 0607, 0640, 0642, 0838, 0902, 1504, 1811, 1812) identified novel potential biomarkers and pathways involved in disease pathogenesis and response to treatment, specifically focusing on the roles of neutrophils, complement, type I interferons, and the pentose phosphate pathway.

Sjogren's Disease

The American College of Rheumatology 2024 Annual Meeting yielded several interesting abstracts on Sjögren's Disease (SjD). Research into the role of syndecan-1 (SDC-1) in SjD pathogenesis (Abstract 0017) revealed that increased SDC-1 expression in a mouse model correlated with increased B-cell chemokine levels, particularly CXCL13. Importantly, heparan sulfate treatment significantly reduced inflammation and chemokine expression, suggesting a potential therapeutic avenue. Further, the discovery of novel TBK1 exosite inhibitors (Abstract L01) offers a promising new therapeutic strategy, demonstrating potent and selective inhibition of the TBK1-STING pathway across cellular and murine models, including a TREX1 knockout mouse model relevant to lupus and SjD, without compromising desirable antiviral or autophagy functions. Finally, research using a large prospective IgG4-RD registry (Abstract 0286) highlighted the high prevalence of organ damage in IgG4-RD patients, particularly in the pancreas and kidneys, emphasizing the importance of long-term monitoring and management. Additionally, a study on the relationship between chronic sinus disease and interstitial lung disease (ILD) in SjD (Abstract 0154) showed a significant association, suggesting that chronic rhinosinusitis might predict the development of ILD in these patients, supporting the idea of a disease continuum between upper and lower airways. The finding that a short-term halt of mycophenolate mofetil (MMF) following herpes zoster vaccination significantly improves the humoral immune response in SjD patients (Abstract 0247) also suggests another avenue to improve vaccination outcomes in this population.

Several abstracts focused on improving diagnosis and management of SjD. A machine learning model (Abstract L11) demonstrated high accuracy in identifying ANA patterns, including the rare DFS pattern, potentially streamlining diagnosis. Studies examining the impact of physical activity (Abstract 0123) and exercise routines (Abstract PP09) on SjD patient outcomes suggested a correlation between increased physical activity and improved quality of life, highlighting the importance of incorporating exercise into treatment plans. The analysis of patient-reported outcomes in inflammatory arthritis and Long COVID (Abstract 0245) revealed a significant negative impact of Long COVID on disease activity and quality of life, suggesting that Long COVID needs to be considered in the management of patients with inflammatory arthritis. Finally, the exploration of novel autoantibodies and biomarkers associated with fetal AVB in high titer anti-Ro exposed pregnancies (Abstract 0444) and the exploration of new candidate biomarkers reflecting changes in CAU activity (Abstract 0383) may lead to improved diagnosis and risk assessment. Abstracts 0775 and 0777 focus on clinical trials and the immunopathogenesis of SjD further clarifying the role of b-cell depletion (Abstract 0775), and the role of different cell populations in lymph nodes versus salivary glands (Abstract 0777). Several other abstracts also provided valuable insights into various aspects of SjD, including genetic architecture (Abstract 0884), cellular heterogeneity in salivary glands (Abstract 0889), autoantigen identification (Abstract 0892), early disease mechanisms (Abstract 0931), the impact of deucravacitinib on patients with SLE and anti-Ro/SSA antibodies (Abstract 0620), and the use of IHC and RNAseq for characterizing and quantifying disease involvement (Abstracts 1409, 1416, 2298, 2299, 2300, 2302).

Psoriatic Arthritis

The ACR 2024 Annual Meeting yielded several interesting abstracts on Psoriatic Arthritis (PsA). Abstract #0067 investigated the role of dendritic cells in the transition from psoriasis to PsA. Researchers used single-cell RNA sequencing to show an increase in type II conventional dendritic cells (cDC2) in psoriasis patients at high risk for developing PsA. These cDC2s exhibited increased expression of CXCR4, a chemokine receptor promoting cell migration, and demonstrated enhanced migration in vitro. This suggests that CXCR4+ cDC2s may play a pathogenic role in the PsO-to-PsA transition and could be targets for preventive therapies. Abstract #0075 explored microRNAs (miRNAs) as potential biomarkers predicting methotrexate (MTX) response in PsA. They identified miR-127-3p as a biomarker for articular response and seven other miRNAs associated with cutaneous response. These findings could personalize MTX treatment, improving outcomes by avoiding ineffective therapies. Abstract #0219 investigated the use of ultrasound to assess nail units in Psoriatic patients, finding significant differences in power Doppler signals between psoriasis patients and controls, suggesting decreased nail bed vascularization might be a marker of psoriatic nail involvement. Finally, abstract #0220 demonstrated that ultrasound-detected lesions of dactylitis, combined with extra-digital inflammation, are predictive of PsA diagnosis in psoriasis patients with finger arthralgia, offering a potential new diagnostic tool.

Several abstracts highlighted the cardiovascular (CV) implications of PsA. Abstract #0141 showed that PsA patients have a higher risk of atherosclerotic cardiovascular disease (ASCVD) than controls, but that treatment with Th17 inhibitors significantly reduces this risk. Abstract #0156 used data from the National Inpatient Sample to show that even in transition-age young adults (18-24 years), PsO/PsA patients were more likely to be admitted for non-fatal CV events than their peers. Abstracts #0205 and #0306 explored the relationship between rheumatic diseases, including PsA, and cardiovascular events (CVE). Abstract #0205 found that PsA had lower CVE rates than rheumatoid arthritis (RA) and systemic lupus erythematosus (SLE), while #0306 did not find a direct correlation between disease activity and increased CV risk in RA or PsA patients using a novel cardiovascular risk calculator. These findings emphasize the importance of considering CV risk assessment and management in PsA patients, particularly given the potential for early CV events in young adults.

Polymyalgia Rheumatica and Giant Cell Arteritis

The American College of Rheumatology 2024 Annual Meeting yielded several interesting abstracts related to Polymyalgia Rheumatica (PMR) and Giant Cell Arteritis (GCA). Abstract #0737 highlighted the significant impact of rurality on GCA diagnosis, finding that rural patients experienced longer delays in diagnosis and increased hospitalizations compared to their urban counterparts. Abstract #0738 explored the use of vascular ultrasound (VUS) in monitoring GCA, demonstrating that VUS quantification scores (Halo Count, Halo Score, and OMERACT GCA US Score) possess both diagnostic and prognostic value. Furthermore, abstract #0741 investigated readily available laboratory tests as markers of PMR disease activity, suggesting serum fibrinogen and haptoglobin as potentially useful indicators for diagnosis and relapse detection. Abstract #0742 assessed the utility of FDG-PET scans for monitoring GCA patients on tocilizumab, concluding that it has limited value during clinical remission. Abstract #0743 evaluated the prognostic significance of FDG-PET/CT findings in PMR, suggesting that increased musculoskeletal inflammation could be a positive prognostic factor. The results of abstract #0770 showed that Upadacitinib 15mg combined with a 26-week glucocorticoid taper regimen demonstrated superior efficacy and reduced glucocorticoid use compared to placebo in GCA patients. Abstract #1640 compared the performance of the new ACR/EULAR 2022 criteria and the ACR 1990 criteria for GCA, showing the new criteria had better sensitivity for large vessel involvement.

Several other abstracts contributed valuable insights. Abstract #0740 characterized the physical phenotypes of aging in PMR patients, finding a high prevalence of frailty and cognitive impairment. Abstract #0744 determined the prevalence and predictive factors of cerebrovascular accidents (CVA) in GCA, identifying visual manifestations as a risk factor. Abstract #0745 assessed the effectiveness of tocilizumab in GCA-aortitis, suggesting that combined therapy may be superior to monotherapy. Abstract #0751 examined sustained drug-free remission in GCA, revealing that it's achieved by a relatively small proportion of patients within a few years of diagnosis. Abstract #0755 identified characteristics at 6 months in new-onset PMR patients associated with prolonged glucocorticoid use. Abstract #0762 explored associations between vascular ultrasound presentation subtypes and ischemic complications in GCA, finding that different patterns of arterial involvement predicted different types of ischemic complications. Abstract #0763 investigated the molecular similarities between histologically positive and negative temporal artery biopsies in GCA, identifying a core set of genes that could inform biomarker development. Abstract #0764 described salvage therapeutic strategies in patients with tocilizumab treatment failure in GCA. Abstract #0849 analyzed the effectiveness and safety of tocilizumab in GCA-aortitis and aneurysms. Abstract #1079 investigated the prevalence of preceding transaminitis in patients with IIM and the timing of CK level checks, highlighting that transaminitis could be an early indicator of IIM. Abstract #1613 explored the relationship between diet and GCA risk, suggesting an association with healthy metabolic profiles. Finally, Abstract #1619 highlighted the need for improved osteoporosis screening and vaccination practices in PMR patients.

Gout

The American College of Rheumatology 2024 Annual Meeting yielded several interesting findings on gout. A first-in-human study (Abstract L05) of PRX-115, a novel PEGylated uricase, showed promising results in lowering serum urate levels for up to 12 weeks following a single intravenous dose, with generally good tolerability. Although one serious adverse event (anaphylactic reaction) occurred, it was resolved, suggesting potential for a wide dosing interval. Another study (Abstract 0133) using NHANES data revealed a significant association between gout and gallstone disease, with individuals having gout showing a 1.6-fold increased odds of also having gallstone disease. This warrants further research to understand the underlying mechanisms, particularly given the shared risk factors and the impact of novel medications like semaglutide and tirzepatide. Regional variations in gout outcomes were also highlighted (Abstract 0142), with the Western US region showing increased length of stay and sepsis rates compared to other regions, potentially influenced by social determinants of health and ethnic differences. Finally, a study (Abstract 0268) demonstrated that a higher rate of weight loss induced by orlistat was associated with lower risks of incident gout and fewer recurrent flares among overweight/obese individuals, supporting the importance of weight management in gout prevention and treatment.

Further studies investigating the long-term safety and efficacy of PRX-115 (Abstract L05) are crucial. The association between gout and gallstones (Abstract 0133) needs further investigation to elucidate the shared pathophysiological pathways and potential implications for management strategies. The observed regional disparities in gout outcomes (Abstract 0142) highlight the need for more individualized treatment plans considering various socioeconomic and ethnic factors. The positive impact of weight loss on gout (Abstract 0268) should prompt increased emphasis on lifestyle interventions, including dietary modifications and weight management programs, as integral parts of comprehensive gout care. The implications of these findings for improving the efficacy and safety of gout management are far-reaching, particularly in optimizing the use of novel therapies like PRX-115 and in addressing the broader health needs of the growing population affected by gout and its associated comorbidities.

Scleroderma

The American College of Rheumatology 2024 Annual Meeting yielded several interesting findings on scleroderma, focusing on biomarkers, treatment response, and disease heterogeneity. A significant finding was the identification of placental growth factor (PlGF) as a promising biomarker for scleroderma renal crisis (SRC) (Abstract 0701). Elevated PlGF levels accurately predicted SRC occurrence and the risk of long-term dialysis, suggesting its potential use in diagnosis and prognosis. Further research into the tolerability of nintedanib in real-world settings revealed common gastrointestinal side effects often leading to medication discontinuation (Abstract 0299), highlighting the need for careful patient selection and monitoring. The study of real-world outcomes of scleroderma patients admitted with acute coronary syndrome (ACS) revealed racial disparities in mortality, with Asians and African Americans facing higher risks (Abstract 0174). Finally, investigation into the prognostic value of Type I interferon (IFN) activity revealed a strong association with a combined morbidity-mortality endpoint in both limited and diffuse cutaneous SSc (Abstract 0695), indicating its potential utility in risk stratification and clinical trial design.

Further research explored the heterogeneity of scleroderma through various approaches. Gene expression profiling identified distinct intrinsic subsets (inflammatory, fibroproliferative, limited, and normal-like) that correlated with longitudinal clinical outcomes (Abstract 0711), suggesting the potential for personalized medicine approaches. A novel unified vascular phenotype (UVP) score, combining peripheral vascular manifestations and FVC/DLCO ratio, demonstrated significant prognostic value for severe vascular events and mortality (Abstract 0694). The study of unique TCR clones in African American patients with systemic sclerosis identified motifs associated with ILD and PAH (Abstract 0958), suggesting a potential role of these clones in driving disease pathogenesis. Finally, the utility of a novel high frequency QUS device demonstrated that it can readily distinguish between SSc and healthy skin based on quantitative assessment of skin microstructure (Abstract 2477), potentially providing a robust quantitative outcome measure for future clinical trials.

Fibromyalgia

The American College of Rheumatology 2024 Annual Meeting yielded several interesting abstracts on fibromyalgia. Abstract #0327 investigated the prevalence of fibromyalgia (FM) in idiopathic inflammatory myopathies (IIM) patients in complete clinical remission, finding a surprisingly high prevalence (40.1%). This suggests that chronic pain in IIM patients, even those in remission, may be attributed to FM and warrants further investigation into the underlying mechanisms and appropriate management strategies. Abstract #1156, using data from the National Databank for Rheumatic Diseases, also reported a high prevalence of comorbid FM in IIM patients (34%), associated with poorer outcomes, poorer socioeconomic factors, and increased healthcare utilization. This underscores the importance of recognizing and managing comorbid FM in IIM patients to improve overall outcomes. Abstract #1178 explored the effects of immersive virtual reality-enhanced meditation (IVRm) on pain in FM patients, finding limited impact on pain but suggesting that IVRm might better reduce awareness of bodily discomfort and negative affect compared to non-immersive meditation. Finally, abstract #1223 examined the prevalence of fibromyalgia in young and middle-aged women in Western India, reporting a prevalence of 6.17%.

Several abstracts focused on the diagnostic and treatment aspects of fibromyalgia and related conditions. Abstract #1219 performed a meta-analysis revealing significantly higher serum calcitonin gene-related peptide (CGRP) levels in fibromyalgia patients compared to controls, suggesting CGRP as a potential diagnostic biomarker. Abstract #1222 explored the use of large language models (LLMs) for fibromyalgia diagnosis, finding that prompt-engineered sentiment analysis could accurately distinguish between fibromyalgia patients and controls based on subtle differences in pain expression. Abstract #1221 investigated the efficacy of immersive virtual reality combined with biofeedback (IVR-BF) for fibromyalgia pain, demonstrating significant improvements in pain intensity and overall impact, although the effect waned over time, suggesting the need for periodic re-treatment. Abstract #1218 presented results from a Phase 3 trial showing that a sublingual cyclobenzaprine significantly reduced fibromyalgia pain and other symptoms, including fatigue and sleep disturbances. The high prevalence of fibromyalgia highlighted in several studies emphasizes the need for further research into its diagnosis, treatment, and management, particularly in diverse populations.